

SUMMARY

1. **Even function:** $f(-x)=f(x)$;
Odd function: $f(-x)=-f(x)$
2. Line $y=ax+b$ (a : the **slope** of the line)

Two lines $y = ax + b, y' = a'x + b'$ are **parallel** $\Leftrightarrow a = a'$ (their slopes are equal)

Two lines $y = ax + b, y' = a'x + b'$ are **orthogonal/perpendicular** $\Leftrightarrow a \cdot a' = -1$ (the product of their slopes is -1)

3. Transformations of functions:

- The graph $y = f(x) + c$ ($c > 0$) is obtained by **shifting** the graph $y = f(x)$ c units **upward**.
- The graph $y = f(x) - c$ ($c > 0$) is obtained by **shifting** the graph $y = f(x)$ c units **downward**.
- The graph $y = f(x - c)$ ($c > 0$) is obtained by **shifting** the graph $y = f(x)$ c units to the **right**.
- The graph $y = f(x + c)$ ($c > 0$) is obtained by **shifting** the graph $y = f(x)$ c units to the **left**.
- The graph $y = cf(x)$ ($c > 1$) is obtained by **stretching** the graph $y = f(x)$ **vertically** by a factor of c .
- The graph $y = cf(x)$ ($0 < c < 1$) is obtained by **compressing** the graph $y = f(x)$ **vertically** by a factor of c .
- The graph $y = f(cx)$ ($c > 1$) is obtained by **compressing** the graph $y = f(x)$ **horizontally** by a factor of c .
- The graph $y = f(cx)$ ($0 < c < 1$) is obtained by stretching the graph of $y = f(x)$ **horizontally** by a factor of c .
- The graph $y = -f(x)$ is obtained by **reflecting** the graph of $y = f(x)$ about the **x-axis**.
- The graph $y = f(-x)$ is obtained by **reflecting** the graph of $y = f(x)$ about the **y-axis**.

4. Limit

A function f is **continuous at** a number a if $\lim_{x \rightarrow a} f(x) = f(a)$

$\lim_{x \rightarrow a} f(x)$ exists $\Leftrightarrow \lim_{x \rightarrow a-} f(x) = \lim_{x \rightarrow a+} f(x)$ (giới hạn trái=giới hạn phải)

5. $y = f(x)$ has a **vertical asymptote** at $x = a$ if $\lim_{x \rightarrow a} f(x) = \infty$ or $\lim_{x \rightarrow a} f(x) = -\infty$.

$y = \frac{p(x)}{q(x)}$ has a vertical asymptote at $x = a$ if $q(a) = 0$.

$y = f(x)$ has a **horizontal asymptote** at $y = a$ if $\lim_{x \rightarrow \infty} f(x) = a$ or $\lim_{x \rightarrow -\infty} f(x) = a$

6. Limit laws

$$1. \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$5. \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

$$3. \lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$$

$$6. \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$$

$$4. \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

7. Differentiation rule

$$(f + g)' = f' + g'$$

$$(f \circ g)' = f'(g) \cdot g'$$

$$(cf)' = c f'$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{gf' - g'f}{g^2}$$

8. The equation of **tangent line** of graph $y = f(x)$ at point (x_0, y_0) is $y - y_0 = f'(x_0)(x - x_0)$
($f'(x_0)$ is the slope of the tangent)

The equation of **normal line** of graph $y = f(x)$ at point (x_0, y_0) is $y - y_0 = \frac{-1}{f'(x_0)}(x - x_0)$

9. Estimate the value of a function at a point: $f(x) \approx f(a) + f'(a) \cdot (x - a)$

Linear approximation of $f(x)$ at $x = a$ is $L(x) \approx f(a) + f'(a) \cdot (x - a)$ (linear approximation is the tangent line)

10. A function f has a **local maximum** (or relative maximum) at c if $f(c) \geq f(x)$ when x is near c .

Similarly, f has a **local minimum** (or relative minimum) at c if $f(c) \leq f(x)$ when x is near c .

11. c is called a critical point of f if $f'(c) = 0$ or $f'(c)$ is undefined.

12. Mean value theorem: If f is continuous on the closed interval $[a, b]$ and f is differentiable on the open interval (a, b) , then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

13. Suppose f'' is continuous near c . If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .

If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

14. f has an inflection point at $x = a$ if $f''(a) = 0$ and f'' change signs at $x = a$.

15. **Newton's method to approximate the solution** of $f(x) = 0$:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

16. Integration rules

$$1. \int_a^b c dx = c(b - a), \text{ where } c \text{ is any constant}$$

$$2. \int_a^b cf(x) dx = c \int_a^b f(x) dx, \text{ where } c \text{ is any constant}$$

$$3. \int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$4. \int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

$$5. \int_a^b f(x) dx = - \int_b^a f(x) dx$$

17. Mean value theorem for integrals (Average value of a function)

If f is continuous on $[a, b]$, then there exists a number c in $[a, b]$ such that

$$f(c) = f_{ave} = \frac{1}{b-a} \int_a^b f(x) dx$$

18. Integration by part

$$\int f(x)g'(x)dx = f(x)g(x) - \int g(x)f'(x)dx$$

19. Numerical integration: From a to b divide n parts: $a = x_0, x_1, \dots, x_{n-1}, x_n = b$

Left endpoint method:

$$\int_a^b f(x)dx \approx \Delta x [f(x_0) + f(x_1) + \dots + f(x_{n-1})]$$

Right endpoint method:

$$\int_a^b f(x)dx \approx \Delta x [f(x_1) + f(x_2) + \dots + f(x_n)]$$

Midpoint method:

$$\int_a^b f(x)dx \approx \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \dots + f(\bar{x}_n)]$$

Trapezoidal method:

$$\int_a^b f(x)dx \approx \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + \dots + 2f(x_{n-1}) + f(x_n)]$$

Simpson's method:

$$\int_a^b f(x)dx \approx \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + f(x_2)] + \dots + \frac{\Delta x}{3} [f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

20. Matrix

$$A + B = B + A \quad AB \neq BA \quad (AB)C = A(BC) \quad A(B + C) = AB + AC$$

$$(A^T)^T = A \quad (kA)^T = kA^T \quad (A + B)^T = A^T + B^T \quad (AB)^T = B^T A^T$$

If $A = A^T$ then A is **symmetric**.

If $AB=BA$ then we say A commutes with B and B commutes with A.

$$A^{-1} = \frac{1}{\det A} \cdot \text{adj } A$$

Let A be a square matrix. Then A is **invertible** $\Leftrightarrow \det A \neq 0$.

$$(A^{-1})^{-1} = A; \quad (AB)^{-1} = B^{-1}A^{-1}; \quad (A^T)^{-1} = (A^{-1})^T$$

Interchange two rows or interchange two columns, $\det A$ changes sign.

Multiply one row (or column) by a nonzero number k, $\det A$ is increased by k times.

Add a multiple of one row (or column) to another row (or column), $\det A$ doesn't change.

$$\det A = \det A^T; \quad \det(A^{-1}) = \frac{1}{\det A}; \quad \det(AB) = \det A \cdot \det B; \quad \det(A + B) \neq \det A + \det B$$

If A is an invertible $n \times n$ matrix then $\det(kA) = k^n \cdot \det A$; $\det(\text{adj } A) = (\det A)^{n-1}$

Theorem: Suppose a system of m equations in n variables $AX=B$ is consistent, and the rank of the augmented matrix is r .

1. The set of solutions involves exactly $n-r$ parameters.
2. If $r < n$, the system has infinitely many solutions.
3. If $r = n$, the system has a unique solution.

Theorem: For homogenous systems: $AX=0$

1. Every homogeneous system **has at least one solution $(0,0,\dots,0)$** , called **trivial solution** (ngiệm tầm thường)
2. If a homogeneous system of linear equations has **nontrivial solution** (ngiệm không tầm thường) then it has infinite family of solutions (vô số nghiệm)
3. If a homogeneous system of linear equations has **more variables than equations**, then it has infinitely many nontrivial solutions.

Theorem: Assume that A is an $n \times n$ matrix. Then

A is **invertible** $\Leftrightarrow$ There exist an $n \times n$ matrix C such that $AC = I_n$.

$\Leftrightarrow A$ can be carried to I_n by elementary row operations.

$\Leftrightarrow$ The homogeneous system $AX=0$ has **only the trivial solution $X=0$** .

$\Leftrightarrow$ The system $AX=B$ has **unique solution** for every choice of column B .

Theorem

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is a matrix transformation induced by the following $m \times n$ matrix

$$A = [T(E_1) \ T(E_2) \ \dots \ T(E_n)]$$

$$\text{where } E_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, E_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, E_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

Theorem 2.6.4

The rotation $R_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the linear transformation with matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

Theorem 2.6.5

Let Q_m denote reflection in the line $y = mx$. Then Q_m is a linear transformation with matrix $\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}$.

Theorem 2.6.6

Let $P_m : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be projection on the line $y = mx$. Then P_m is a linear transformation with matrix $\frac{1}{1+m^2} \begin{bmatrix} 1 & m \\ m & m^2 \end{bmatrix}$.

Theorem

If the matrix of T is A and the matrix of S is B then the matrix of $S \circ T$ is BA .

How to find eigenvalues and eigenvectors?

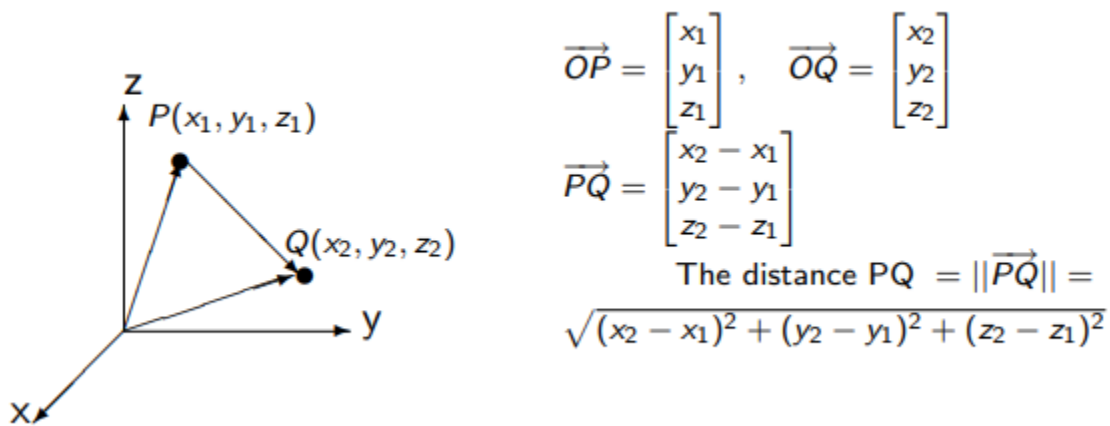
Let A be a square matrix.

- The **characteristic polynomial** of A is $c_A(x) = \det(xI - A)$.
- Eigenvalues of A are the solutions of the equation $c_A(x) = 0$.
- To find λ -eigenvectors, solve the system $(\lambda I - A)X = 0$.

Diagonalization Algorithm. Let A be an $n \times n$ matrix.

- Find eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A .
- For each eigenvalue compute basic eigenvectors.
- If there are exactly a total of n basic eigenvectors, then A is diagonalizable.
- In this case D is the diagonal matrix with $\lambda_1, \lambda_2, \dots, \lambda_n$ in the diagonal, and P is the matrix whose columns are the basic eigenvectors.

21. Vector



Equation for the line passing through the point $P_0(x_0, y_0, z_0)$ and **parallel** to the vector $\vec{d} = (a, b, c)$ is

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases} \quad (t \in \mathbb{R})$$

Equation for the plane passing through the point $P_0(x_0, y_0, z_0)$ and **orthogonal** to vector $\vec{n} = (a, b, c)$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

The dot product

Let $\vec{u} = [u_1 \ u_2 \ u_3]^T$, $\vec{v} = [v_1 \ v_2 \ v_3]^T$. The **dot product** of $\vec{u}$ and $\vec{v}$, is

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$$

Properties

- $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
- $(k\vec{u}) \cdot \vec{v} = k(\vec{u} \cdot \vec{v}) = \vec{u} \cdot (k\vec{v})$
- $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
- $\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}}$
- Let θ be the angle between $\vec{u}$ and $\vec{v}$, then $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}$
- $\vec{u}$ and $\vec{v}$ are orthogonal if and only if $\vec{u} \cdot \vec{v} = 0$

Projection of $\vec{u}$ into $\vec{v}$: $\text{proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v}$

The **cross product** of two vectors $\vec{v} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ is the vector

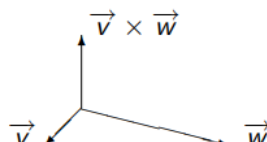
$$\vec{v} \times \vec{w} = \left[\begin{vmatrix} y_1 & y_2 \\ z_1 & z_2 \end{vmatrix}, - \begin{vmatrix} x_1 & x_2 \\ z_1 & z_2 \end{vmatrix}, \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} \right]^T$$

Theorem

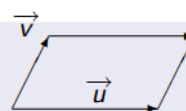
If $\vec{u} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$ then $\vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \\ z_0 & z_1 & z_2 \end{vmatrix}$.

Note.

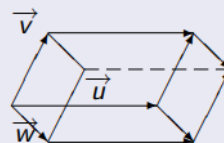
The cross product $\vec{v} \times \vec{w}$ is orthogonal to both $\vec{v}$ and $\vec{w}$.



The area of the parallelogram determined by $\vec{u}$ and $\vec{v}$ is $\|\vec{u} \times \vec{v}\|$.



The volume of the parallelepiped determined by $\vec{u}$, $\vec{v}$ and $\vec{w}$ is $|\vec{u} \cdot (\vec{v} \times \vec{w})|$.



Note. If $\vec{u} = [x_0 \ y_0 \ z_0]^T$, $\vec{v} = [x_1 \ y_1 \ z_1]^T$, $\vec{w} = [x_2 \ y_2 \ z_2]^T$ then the volume of the parallelepiped determined by $\vec{u}$, $\vec{v}$ and $\vec{w}$ is the *absolute*

value of the determinant of the matrix $\begin{bmatrix} x_0 & x_1 & x_2 \\ y_0 & y_1 & y_2 \\ z_0 & z_1 & z_2 \end{bmatrix}$

A subset U of $\mathbb{R}^n$ is called a **subspace** of $\mathbb{R}^n$ if:

- $\vec{0} \in U$
- (Closed under addition) $\vec{u}, \vec{v} \in U \implies \vec{u} + \vec{v} \in U$
- (Closed under multiplication by scalars) $\vec{u} \in U \implies k\vec{u} \in U$.

Note: Subspaces of $\mathbb{R}^3$ are lines passing through the origin or planes passing through the origin.

The set $\{\vec{X}_1, \vec{X}_2, \dots, \vec{X}_k\}$ is called **linearly independent** if, whenever $t_1\vec{X}_1 + t_2\vec{X}_2 + \dots + t_n\vec{X}_k = 0$ then $t_1 = t_2 = \dots = t_k = 0$.

Let U be a subspace of R^n . A set $\{\vec{X}_1, \vec{X}_2, \dots, \vec{X}_m\}$ is called a **basis** of U if:

- $\{\vec{X}_1, \vec{X}_2, \dots, \vec{X}_m\}$ is linearly independent.
- $U = \text{span}\{\vec{X}_1, \vec{X}_2, \dots, \vec{X}_m\}$

- A set $\{X_1, X_2, \dots, X_k\}$ of vectors in R^n is called an **orthogonal** set if $X_i \neq 0$ for all i , and $X_i \cdot X_j = 0$ for all $i \neq j$.
- A set $\{X_1, X_2, \dots, X_k\}$ is called an **orthonormal** set if it is orthogonal and $\|X_i\| = 1$ for all i .

- The **column space** of a matrix A , denoted by $\text{col}(A)$, is the subspace spanned by the columns of A .
- The **row space** of a matrix A , denoted by $\text{row}(A)$, is the subspace spanned by the rows of A .

Let $A \xrightarrow{\text{Echelon}} R$. Then:

- The rows of R containing the leading 1 are a basis of the row space of A .
- The columns of A corresponding to the columns of R that contain the leading 1 are a basis of the column space of A .

Theorem

Let A be an $m \times n$ matrix. Suppose $\text{rank}(A) = r$. Then:

- $\dim(\text{row}(A)) = \dim(\text{col}(A)) = \dim(\text{im}(A)) = r$
- $\dim(\text{null}(A)) = n - r$

